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El nino model equation containing gravity

1 message

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In El Niño modeling, the primary mathematical framework that explicitly incorporates gravity is the **Reduced-Gravity Shallow-Water Equation** system. This system forms the foundation of intermediate-complexity models, such as the famous **Cane-Zebiak model**, used to track thermocline variations and ocean dynamics during El Niño Southern Oscillation (ENSO) events. [1, 2, 3]

The Reduced-Gravity Shallow-Water Model Equations

The model simplifies the tropical Pacific Ocean into a dynamic, warm upper layer of average depth H and density ρ floating over a deeper, motionless cold layer of slightly higher density $\rho + \Delta\rho$. [1]

The system is governed by three primary partial differential equations on an equatorial β -plane: [1]

$$\frac{\partial u}{\partial t} - fv = -g' \frac{\partial h}{\partial x} + \frac{\tau^x}{\rho H}$$

$$\frac{\partial v}{\partial t} + fu = -g' \frac{\partial h}{\partial y} + \frac{\tau^y}{\rho H}$$

$$\frac{\partial h}{\partial t} + H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0$$

Parameter Definitions

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- h : The anomaly of the thermocline depth (how much the boundary layer fluctuates).
- u, v : The zonal (east-west) and meridional (north-south) ocean current velocities.
- τ^x, τ^y : Zonal and meridional wind stress forcing at the sea surface.
- $f = \beta y$: The Coriolis parameter, which varies with latitude from the equator.
- g' : The **Reduced Gravity** parameter. [1]
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The Role of Gravity (g')

True gravitational acceleration ($g \approx 9.81 \text{ m/s}^2$) is too intense for modeling internal ocean layers because buoyancy counters most of its effect. Instead, El Niño models use **reduced gravity (g')**, calculated as: [1, 4]

$$g' = g \frac{\Delta \rho}{\rho}$$

Because density variations ($\Delta \rho$) between the warm pool and the cold deep ocean are miniscule, g' is typically reduced to roughly 0.03 m/s^2 .

Why Gravity Matters for El Niño

In these equations, gravity functions as the vital restoring force: [5]

1. **Wave Propagation:** Reduced gravity dictates the speed of internal **equatorial Kelvin waves** ($c = \sqrt{g'H}$). When Pacific trade winds weaken during an El Niño onset, gravity pulls the elevated western Pacific warm pool down, sending Kelvin waves traveling eastward along the equator to depress the thermocline near South America. [6, 7]
2. **Pressure Gradients:** The terms $-g' \frac{\partial h}{\partial x}$ and $-g' \frac{\partial h}{\partial y}$ represent horizontal pressure gradient forces. They describe how differences in water height push sea currents across the Pacific Basin. [8, 9]

Would you like me to show you how these fluid equations are coupled to the **atmospheric wind stress equations** to create a complete **delayed oscillator loop**, or do you want to explore the numerical **Kelvin wave speed** calculations?

- [1] <https://www.nature.com>
 [2] <https://horizon.documentation.ird.fr>
 [3] <https://journals.ametsoc.org>
 [4] <https://en.wikipedia.org>
 [5] <https://medium.com>
 [6] <https://science.nasa.gov>
 [7] <https://physicsworld.com>
 [8] <https://dbarker.uk>
 [9] <https://personalpages.manchester.ac.uk>

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